

Nishant Bakshi

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Electrical engineer with 2+ years in semiconductor reliability and test engineering, plus an MSEE from USC in analog/mixed-signal IC design. Skilled in circuit simulation, PCB design, and hardware development, with 10+ projects spanning CMOS chips, power electronics, and audio systems.

TECHNICAL SKILLS

Circuit Simulation: Cadence Spectre, LTSpice, PySpice, PLECS

Circuit Layout: Cadence Virtuoso, Altium, KiCAD

Programming: Python, Embedded C, C/C++, JSL, LabVIEW, Git

Laboratory Equipment: Probe Station, Oscilloscope, Parametric Analyzer, Curve Tracer, Function Generator

Audio & Acoustics: VituixCAD, Akabak, REW, WinISD, Boundary Lab, MaxMSP

Manufacturing & Fabrication: SMD/Through-Hole Assembly, PCB Fabrication, CNC Machining, Wire Bonding

EXPERIENCE

Product & System Designer

Jan. 2026 – Present

Common Wave Hi-Fi

Los Angeles, CA

- Designed 3 complete loudspeaker enclosure and crossover systems using BEM/LPM simulation (VituixCAD, Akabak, Boundary Lab), validated through gated and ground-plane acoustic characterization (REW, Audacity)
- Built a custom impedance analysis test fixture; use oscilloscopes and function generators to tune crossovers
- Led rapid-prototyping cycles from concept to production-ready design, compressing cabinet builds and crossover iterations into roughly 6 weeks per speaker; targeting an initial 20–30 unit production run once a manufacturer is selected
- Repaired 20+ vintage tube and transistor amplifiers and turntables, averaging 1-week turnaround against a standing 6-week backlog, including tube rebiasing and component-level resoldering

Graduate Research Assistant

May 2025 – May 2026

IEDM Group, USC

Los Angeles, CA

- Diagnosed and characterized a custom 64-channel, 640MHz S/H chip using probe stations, parametric analyzers, oscilloscopes, and an off-board ADC, contributing to a publication in IEEE Solid-State Circuits Letters
- Assembled, reworked, and brought up mixed-signal PCBs for chip validation, including SMD soldering and signal debugging
- Designed a PGA-socketed S/H test PCB and wirebonding diagram for the custom CMOS chip; interfaced with packaging vendors for chip assembly

Maker Space Student Worker

Aug. 2024 – May 2026

Baum Family Maker Space, USC

Los Angeles, CA

- Wrote curriculum and personally led 2 of 4 weekly soldering certification sessions (2 hrs each), training an estimated 250+ students over 4 semesters
- Provided PCB design review, electronic debugging, and manufacturing support to 6+ USC design teams, including Formula SAE, Autonomous Underwater Vehicle, Solar Car, and Concrete Canoe
- Owned uptime and calibration for the lab's electronics bench, including oscilloscopes, function generators, power supplies, and soldering stations, keeping test equipment mission-ready for 6+ design teams
- Managed consumable parts inventory and scheduling for the soldering certification program

Engineer II - Device

Sep. 2022 – Aug. 2024

Microchip Technology Inc.

Colorado Springs, CO

- Developed C++/Python/LabVIEW stress test programs for TDDB, NBTI, HCI, and Vramp reliability testing
- Authored 3 reliability qualification reports for 180nm+ CMOS technologies, supporting JESD47-based industrial qualification and AEC-Q100 automotive-grade qualification for two technologies
- Characterized L-dependent Pelgrom mismatch on 6,500+ halo-implanted CMOS devices across 4 temperatures (26,000 IV Sweeps), supporting analog design teams
- Engineered secure wireless tester-to-server data transfer, eliminating USB-based security risks
- Developed JSL data analysis scripts in JMP, replacing manual Excel-based calculations and reducing reporting time from 2 weeks to 2–4 days

- Developed a helium-bomb autoclave test monitoring hermetic seal integrity of bonded BAW resonator wafers
- Spotted inconsistent reliability qualification methodology across 5 internal labs and proposed changes to bring testing back into alignment

Undergraduate Research Assistant (Volunteer)

May 2021 – Dec. 2021

NYU NanoLab

New York, NY

- Diagnosed and repaired a corrupted Cadence Virtuoso design environment for a legacy 0.25μm CMOS biosensor project, resimulating op-amp and voltage reference circuits to match prior progress reports
- Evaluated a node migration path for 0.25μm to 10nm CMOS for a future biosensor chip redesign
- Designed and simulated a bandgap voltage reference circuit in Cadence Virtuoso to validate simulation methodology

EDUCATION

University of Southern California

Aug. 2024 – May 2026

Master's of Science in Electrical Engineering

Los Angeles, CA

New York University

Aug. 2018 – May 2022

Bachelor's of Science in Electrical Engineering

New York, NY

PROJECTS

RFID Tool Lockout System | Embedded C, KiCAD, SMD Assembly

Aug. 2025 – Present

Work Project — USC Maker Space

- Designed an ESP32 + PN532-based RFID access control system replacing shared passwords and lockout codes with university ID swipe access, deployed on 2 laser cutters and 1 waterjet cutter in the USC Maker Space
- Developed ESP32 firmware in ESP-IDF (C/C++) for HID emulation and RFID UID reading; designed a custom carrier PCB and 3D-printed enclosure for the dev board assembly
- Develop a cloud registration and staff-approval pipeline with wireless UID database sync to each device
- Lead a lab-wide rollout to all Baum Family Maker Space users and design a custom PN532 reader circuit to replace the dev board

1T1R Neuromorphic RF Classifier | Analog Design, LTSpice, Altium, SMD Assembly

May 2025 – Present

Research Project — IEDM Group, USC

- Collaborated with 2 PhD students to design a board-level 1T1R neuromorphic RF classifier
- Designed and simulated a unity-gain 15MHz TIA with programmable feedback network
- Laid out and assembled the TIA and S/H debug PCBs in Altium Designer
- Validated circuit performance on the bench against simulations

250W 2-Switch Forward Converter | PLECS, Power Converter Design, Magnetics Design

Mar. 2026 – May 2026

Class Project

- Designed a 250W, 450VDC-to-190VDC two-switch forward converter with custom transformer and inductor magnetics
- Selected Wolfspeed SiC MOSFETs and Schottky diodes for medium-voltage switching, and designed a Type-II compensator and feedback network for output regulation
- Simulated power stage and control loop performance in PLECS, achieving 92.6% efficiency at 300W with sub-2% output ripple during load transients

7W Class-A Power Amp | Analog Circuit Design, KiCAD, Through-Hole Assembly

May 2025 – Sep. 2025

Personal Project

- Designed a single-ended Class-A amplifier with NFET output, active load, JFET driver, and BJT current source
- Laid out a through-hole PCB in KiCAD, and hand-assembled the board
- Integrated the amplifier into a heatsink-equipped chassis for thermal management of the Class-A bias current

Pseudo-Stereophonic Up-Mix System | MaxMSP for Live, DSP, Psychoacoustics

Apr. 2025 – May 2025

Class Project

- Led development of a 2-to-3 channel (2PH3) pseudo-stereophonic up-mixing system in MaxMSP for Live, avoiding the -6dB signal loss of traditional resistor-based up-mix circuits
- Drove 3 DAC and power amplifier channels to enable real-time A/B switching between stereo and 2PH3 playback

- Ran a listening study with nearly 25 participants, finding 2PH3 improved center-channel localization and listener envelopment at a measured trade-off in apparent source width

12 Bit 10MS/s Asynchronous SAR ADC | *Cadence Spectre, Mixed-Signal Design* Mar. 2025 – May 2025
Class Project

- Designed a 12-bit, 10MS/s asynchronous SAR ADC in TSMC 45nm CMOS, including the differential CDAC, strongarm comparator, D-flip-flops, and self-timed asynchronous control logic
- Implemented monotonic switching and bottom-plate sampling to reduce pedestal error and eliminate static power consumption
- Simulated performance in Cadence Spectre and recovered waveforms via a custom Python script, achieving a 9.03-bit ENOB (56.1 dB SNDR) at 1MHz
- Developed a SPICE-netlist prompting method for AI-assisted transistor sizing (ChatGPT o3) as a professor-assigned exercise, producing one of the few functional AI-sized designs among 15 class teams

180nm 5.7GHz Rail-to-Rail OTA | *Cadence Virtuoso/Spectre, Analog IC Design* Oct. 2024 – Dec. 2024
Class Project

- Designed a 180nm CMOS rail-to-rail folded cascode OTA in Cadence Spectre, achieving 51 dB DC gain, 5.7 GHz unity-gain bandwidth, and 0.77 nV/ $\sqrt{Hz}$ input-referred noise
- Designed a Differential-Difference Amplifier (DDA) based common-mode feedback circuit, adapted from IEEE literature, to set output common-mode voltage without poly resistors
- Standardized device sizing in practical multiples (e.g., 18 μ m increments) to align with Process Control Monitoring structures, applying wafer-level characterization experience from Microchip

Mini Onken Speaker | *Fusion360, VituixCAD, WinISD, LTSpice* Jan. 2024 – May 2024
Personal Project

- Designed an 8 cubic foot scaled-down Onken enclosure with an external 811 sectoral horn for vintage 1970s Altec drivers, using lumped-parameter modeling in VituixCAD and WinISD
- Fabricated the enclosure from marine-grade Douglas fir plywood using CAD/CAM modeling in Fusion 360 and CNC routing
- Designed, built, and bench-measured a crossover network based on the Altec Model 19 topology

Pocket Photometer | *OrCAD Capture, KiCAD* May 2021 – Dec. 2021
Volunteer Research — NYU MicroParticle Photo-Physics Lab

- Contributed to the development of a battery-powered optical power meter for the Pocket Optics Kit (PO-1), a take-home lab kit now sold commercially on Amazon
- Simulated the TIA circuit in OrCAD Capture, laid out the PCB in KiCAD, and bench-tested the assembled board
- Validated 4 power ranges (1 μ W - 1mW) across a 350-1050nm wavelength range, achieving 40-75% higher accuracy than smartphone light-meter apps

BioFET for DNA Rapid Detection | *Cadence Virtuoso, Altium, Analog IC Design* May 2021 – Nov. 2021
Volunteer Research — NYU NanoLab

- Reconstructed and resimulated legacy TIA, OTA, BGVR, and PGA circuits for a CMOS DNA biosensor chip after the original Cadence files were found irreparably corrupted, matching results to prior documentation
- Designed the chip wire-bonding board and characterization PCBs in Altium Designer
- Authored a final report documenting standard operating conditions and design methodology to support future resizing of the chip from TSMC 0.25 μ m to 10nm

PUBLICATIONS

- M. Hamada, M. Toubar, Z. Wang, **N. Bakshi**, H.-C. Chen, M. Palaria, J. Yang, and M. S.-W. Chen, “A 1.2-V, 8.5-ns Settling, Bi-Slewing CMOS Driver for Heavy Loads in Discrete-Time Analog Computing Applications,” *IEEE Solid-State Circuits Letters*, in press.